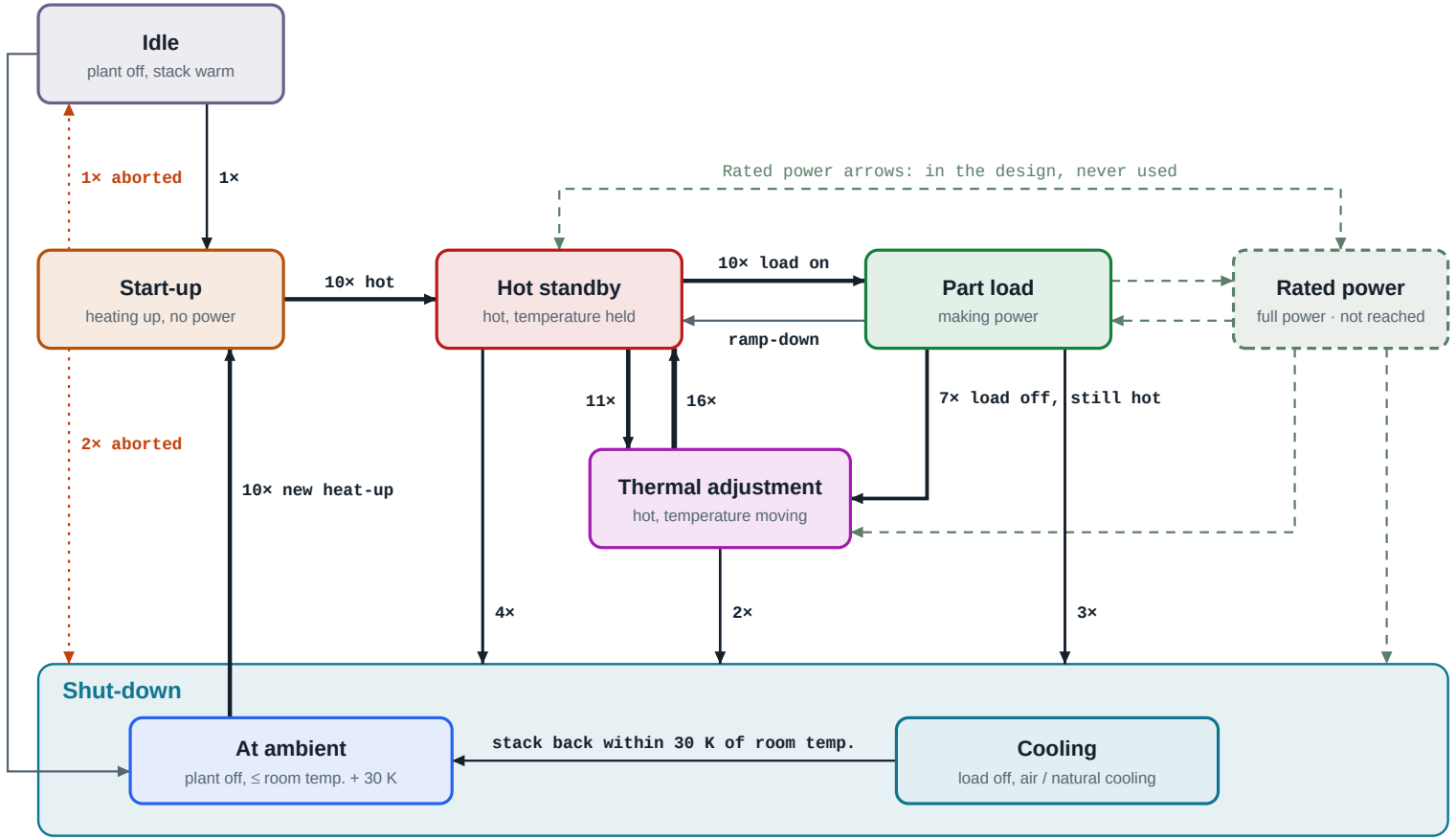
Operating stage map

The plant always moves around one loop: **Shut-down** → **Start-up** → **Hot standby** → **making power** → **back to Shut-down**. Every minute of the data gets one of seven stage labels, and only certain moves between them are allowed. The synthetic data must follow the same loop.

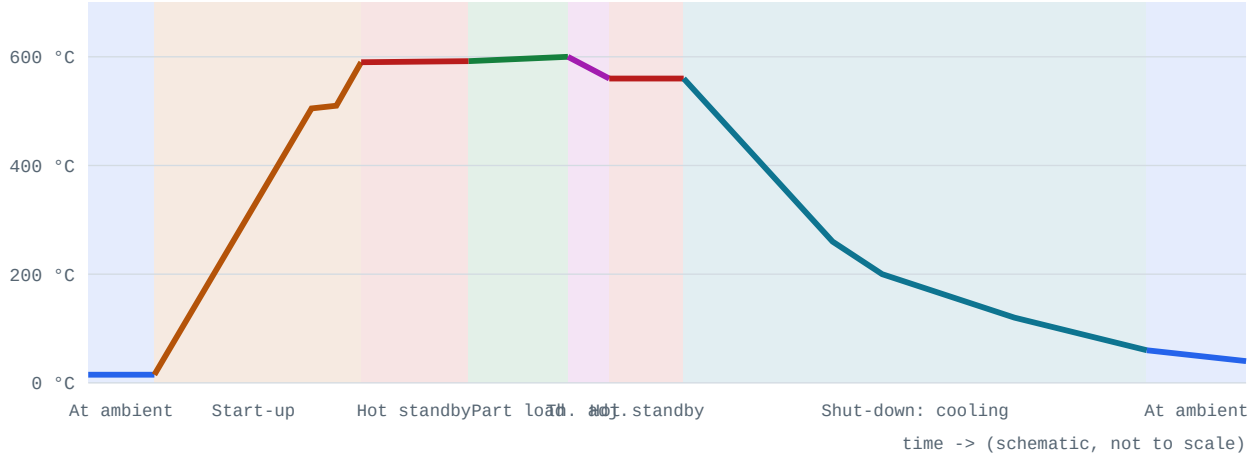


Solid = allowed and used (thicker = more often) · thin grey = allowed, not used yet · green dashed = rated power, never reached · orange dotted = seen in the data, outside the rules. Numbers = times seen in 39 days (90 stage periods).

If you remember one thing: the plant runs one loop, cold → heat-up → hot and steady → power → cool-down, and power can only start from “hot and steady”.

The seven stages

Each stage is told apart by two questions: **is the stack making power?** and **is its temperature rising, steady or falling?**



Schematic stack temperature through one campaign (not to scale in time).

Stage	What it means	Power / temperature	How the data spots it	From → to
Idle	The plant is switched off but the stack is still warm, and no shut-down came before it. It is a label for the data, not a real plant stage. Seen only in the first data file (30 Oct 2025).	0 Warm, slowly drifting (about –1 to –2 °C per hour)	No current, heater off, air off, temperature changing less than 10 °C/h, more than 30 K above room temperature, no Shut-down before it.	from: Start of the data to: Start-up, Shut-down
Start-up	Heating the stack from room temperature until it is hot enough to make power. The overnight pause on 17/18 March 2026 counts as part of one Start-up (“heat-up interrupted”).	0 Rising, room temperature → operating range (about 50 °C/h; limit 120 °C/h)	No current, temperature rising faster than 10 °C/h, stack not yet hot since the last Shut-down.	from: Shut-down (or Idle, once) to: Hot standby
Hot standby	Hot and ready, making no power, temperature held steady. It is the only door into making power: the last 15 minutes before load are always Hot standby (“pre-load”).	0 Constant: 575–610 °C before load, or a long hold at about 510 °C	Current below 1 A, temperature changing less than 5 °C/h, and either at least 550 °C, or 450–550 °C held for 6 hours or more (low-temperature hold).	from: Start-up, Thermal adjustment, Part load to: Part load, Thermal adjustment, Shut-down
Thermal adjustment	The stack is hot and making no power, but its temperature is being moved on purpose: up (re-heat) or down (cool-back), for example from the 510 °C hold to 590 °C.	0 Rising or falling inside the hot range (above 450 °C)	Current below 1 A, above 450 °C, temperature changing 5 °C/h or more, after the stack has been hot.	from: Hot standby, Part load to: Hot standby, Shut-down
Part load	The stack is making electricity, but less than full power. All load in the data is part load: at most 16.2 A per string and 33.6 kW.	Between 0 and full power Operating range	Current per string at least 1 A and below 30 A.	from: Hot standby to: Hot standby, Thermal adjustment, Shut-down
Rated power	Full design power: 50 kW AC / 60 kW DC, 30 A per string. The plant never got there in the data, so its arrows are in the design but have never been used.	50 kW AC / 60 kW DC Operating range	Current about 30 A per string.	from: Hot standby, Part load to: Part load, Hot standby, Thermal adjustment, Shut-down
Shut-down	Everything from switching the load off until the next heat-up. It has two parts: Cooling (load to zero, cooling with air, then natural cooling once the air is off) and At ambient (plant off, stack back near room temperature, waiting).	Falling to 0 Falling: about –45 to –75 °C/h with air (March), –1 to –2 °C/h without	Current going to zero, then cooling faster than 10 °C/h, or air on while cooling, or natural cooling after it. Within 30 K of room temperature with the plant off = At ambient.	from: Hot standby, Thermal adjustment, Part load, Idle to: Start-up

Room temperature is not measured directly: the middle value of four supply-line probes (usually 10–18 °C) is used, so “within 30 K” means below about 40–48 °C.

If you remember one thing: “Hot standby” means hot *and steady*. A hot stack whose temperature is moving, with no power, is a Thermal adjustment.

Moving between stages

Of the 42 possible moves between seven stages, **18 are allowed**. In the data, 10 of them happened, 74 times in total; 2 moves outside the rules happened 3 times.

from ↓ to →	Idle	Start-up	Hot standby	Th. adjust.	Part load	Rated power	Shut-down
Idle	–	1					0
Start-up	1	–	10				2
Hot standby		X	–	11	10	0	4
Th. adjust.			16	–			2
Part load		X	0	7	–	0	3
Rated power			0	0	0	–	0
Shut-down		10			X		–

Green = allowed (number = times seen; pale = allowed, not seen) · orange = outside the rules but seen · X = technically impossible · blank = not allowed. Rated power was never reached.

The plant's ground rules

1. **A Start-up only follows a Shut-down** (not Hot standby, not Part load). Exception: Idle in the very first file.
2. **Part load only follows Hot standby**; never straight from Shut-down.
3. **Hot standby means steady**; a hot, unloaded stack changing ≥ 5 °C/h is in Thermal adjustment.
4. **A pause in a heat-up is still the same Start-up** (17/18 March 2026, plant off overnight, 188 → 167 °C).

Still open

Three heat-ups stopped before the stack got hot: Start-up → Shut-down (13 Nov 2025, 30 Jan 2026) and Start-up → Idle (30 Oct 2025). **Decision needed:** add an “aborted start-up” arrow, or keep them flagged? Also open: the 30 K “near room temperature” margin (40 K would move 19 h on 26 and 28 March from cooling to at-ambient).

If you remember one thing: two doors only: heating up starts after a Shut-down, and power starts after Hot standby.